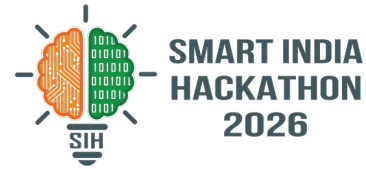


SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – SIH26097

Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY

Theme – Agriculture, FoodTech & Rural Development

PS Category – Software

Team ID – _____ (fill from the SIH portal)

Team Name (Registered on portal) – Cognify



SMART INDIA HACKATHON 2026



THE PROBLEM

A form picks the course, not the person

so SC job seekers land in training that was never going to fit them.

41%of certified trainees got placed
CAG 2025 [S5]**51.6%**of rural women have no phone of their own
NSO 2025 [S2]**66.1%**SC literacy: forms fail many people
Census 2011 [S1]THE SOLUTION *one free call on any keypad phone, guided end to end*

1

Missed call, free callback*from a govt 1600 number**

Within a minute, in the caller's language. No app, no internet, no reading.



2

Keys + own words*keys for facts, voice for skills*

Age, education, distance by keypad; then they describe their work.



3

AI builds a full profile*read back; caller presses 1*

Skills mapped to NCO codes and NSQF qualifications. No paper.



4

3 pathways that fit*each with a plan + money path*

Training, RPL certificate, apprenticeship or own work, near home.



5

Stays till they earn*SMS, reminders, follow-ups*

Case ID + documents by SMS; calls at 1, 3 and 6 months.

WHY IT WORKS WHERE A FORM FAILS *six things only HunarMarg does***Any keypad phone**

missed call, free, in their own language

**Never invented**

every fact from govt data, checked before it is spoken

**Writes the profile itself**

officers only review and approve

**Money path**

which loan or grant fits, and the monthly EMI

**Outcomes that can't be faked**

random-digit liveness check + employer check

**Learns from verified jobs only**

fake placements can't teach it



Technical Approach

Indian, open-source models, fine-tuned by us, running on Indian servers



TECH STACK *Indian and open-source, fine-tuned by us*

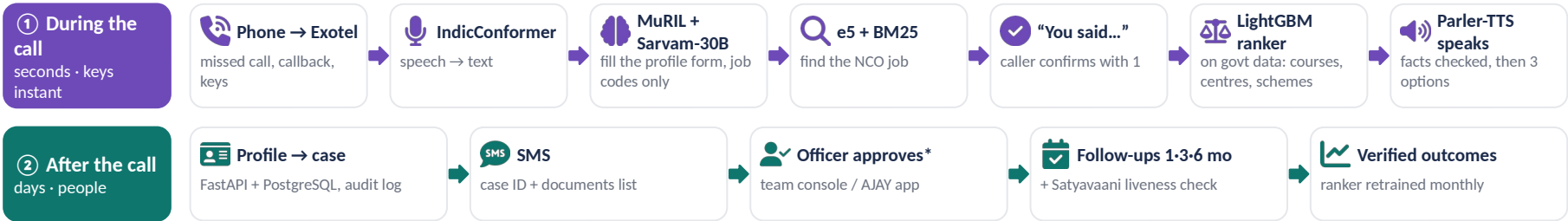
Calls & messages
Exotel · 1600 number* · SMS

AI models (fine-tuned)
IndicConformer · Indic Parler-TTS · Sarvam-30B · MuRIL · e5 · LightGBM · Satyavaani

Backend & data
+ pgvector · Indian GPU cloud

Apps & govt links
React console · NCO-2015 · NQR · DigiLocker* · AJAY app*

HOW IT WORKS *two lanes: during the call (AI, seconds) and after it (people, days)*



AI understands → data and rules decide → facts checked → humans approve GPU down → Sarvam API · AI stuck → human counsellor

OUR MODELS *what each one does, how we train it, the bar it must pass*

IndicConformer-600M <i>Hindi error ≤ 20%</i> Works: 22 languages; 8 kHz phone audio → text Trains: 100 h consented calls per language + IndicVoices · ~72 GPU-h	Indic Parler-TTS <i>listeners ≥ 4/5</i> Works: 21 languages; text + "calm, clear voice" → speech Trains: 12 h of one local voice artist per language · ~24 GPU-h	Sarvam-30B <i>right job in top 3 ≥ 90%</i> Works: fills a fixed form: job codes, years, skills; never advises Trains: LoRA on 5,000 checked story → form pairs · ~16 GPU-h
multilingual-e5 + BM25 <i>right job in top 5 ≥ 90%</i> Works: keyword + meaning score → top 20 → reranker → top 3 Trains: every "1" at read-back = right pair, "none" = wrong pair	LightGBM LambdaRank <i>gender gap ≤ 5 pts</i> Works: filters, then 6 factors: wish, skills, demand, reach, finish, pay Trains: officer-set weights (AHP) first; then verified jobs at 6 months only	Satyavaani (team's own) <i>flags, never rejects</i> Works: 4 random digits + fake-voice score; no voiceprint stored Trains: Indian phone speech + 3 voice-cloning tools

Feasibility and Viability

Can it be built and run? Yes: every part already runs in India, and it costs 0.24% of GIA per person.

ALREADY WORKING IN INDIA *nothing here is untested*

22 crore

feature-phone users

The channel already exists.

Business Standard 2026 [S11]

3 crore+

reached by Kilkari

A govt voice-call line; answers rise 50% → 76% by the 3rd try.

BMJ Global Health [S12]

22 · 21

languages, open models

Indian speech (IndicConformer) and voice (Parler-TTS) models exist.

AI4Bharat [S15]

19.3%

speech error to beat

Best Indian speech AI on IndicVoices; we switch only after beating it.

Sarvam 2026 [S14]

Prototype

works on real calls

3 languages, profile + read-back, top 3 from 115 courses, 124 centres.

GitHub [S21]

WHAT IT COSTS *four final numbers; every rate is in the calculations*

₹163

per person, all-in
at 50,000 people a year

0.24%

of GIA support per person
₹163 of ₹69,200 [S3]

₹8.8 L

a year of GPUs in India
2× L4 + 1× L40S [S17]

₹2 L

one time per language
100 h data + voice + GPU

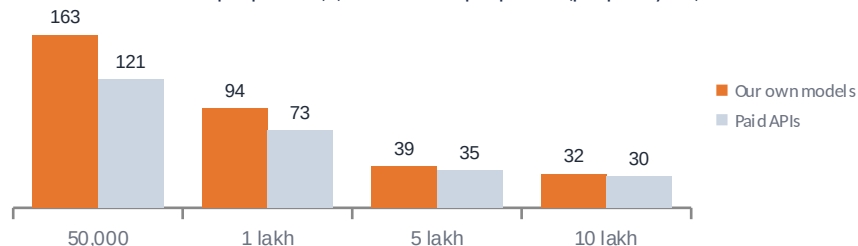
Collect 100 h

Fine-tune ~112 GPU-h

Beat 19.3%

Improve monthly

Cost per person (₹) falls as more people call (people a year)



WHAT COULD GO WRONG, AND WHAT ALREADY HANDLES IT

✗ **Village dialects** ✓ keys for facts, every answer read back, human fallback

✗ **Unknown numbers ignored** ✓ they call first; callback from an official 1600 number*

✗ **AI says something wrong** ✓ only facts that match a govt data row are spoken

✗ **Our model worse than APIs** ✓ switch a language only after it beats 19.3%; API backup

Who gains? SC families, officers, training centres and the Ministry.



WHAT CHANGES *four numbers, each with its source*

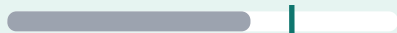
placed after training



CAG found 41% placed; 70% is the GIA target. Courses that fit close the gap.

CAG 2025 [S5] · PM-AJAY [S4]

stay in the job



when trainees hear real pay and place before joining; HunarMarg says it on the call.

J-PAL field trial [S13]

officer time saved per person

≈ 52 staff-years a year: the AI writes the profile, officers approve.

our estimate

saved on one loan

on a woman's ₹80,000 tailoring unit, with the right grant + loan path.

our calculation



~50,000 GIA beneficiaries a year [S3] **47,000+** SC-majority villages already on the PM-AJAY dashboard [S10]

WHO GAINS WHAT *social · economic · governance · environment*



SC families, women, non-readers

Right course the first time, free, in their own language; women-only batches; calls until they earn.



the Ministry and the trainee

0.24% extra per person makes sure the other 99.76% buys the right course; the cheapest loan path.



officers and CSC operators

Ready profiles, one case record from call to job, block-level demand for the annual plan.



every district office

No paper forms, fewer trips to offices, runs on Indian cloud.

Every number in this deck, with where it came from

SOURCES *by type, each with its website*



Government data

- S1 Census 2011 (MoSJE Handbook 2021): SC literacy 66.1% · [socialjustice.gov.in](#)
- S2 NSO Telecom survey 2025: phone ownership · [mospi.gov.in](#)
- S3 MoSJE PM-AJAY factsheet, 28 Sep 2026: ₹1,730 cr, 2.5 lakh people · [piib.gov.in](#)
- S4 PM-AJAY guidelines: 70% placement, 30% women · [socialjustice.gov.in](#)
- S5 CAG Report No. 20 of 2025: 41% placed · [cag.gov.in](#)
- S6 NCO-2015 · S7 NQR / NCVEIT · S8 NSFDC · S9 PIB PM Vishwakarma
- S10 PIB: PM-AJAY portal + AJAY app, 26 May 2026 · [piib.gov.in](#)



Research

- S11 Business Standard, Apr 2026: 22 crore feature-phone users
- S12 BMJ Global Health: Kilhari reach and answer rates · gh.bmj.com
- S13 Chakravorty et al., J-PAL: job information → retention 62% → 73% · povertyactionlab.org



Technology

- S14 Sarvam: Saaras v3, 19.31% WER on IndicVoices · [sarvam.ai](#)
- S15 AI4Bharat: IndicConformer, IndicVoices, Indic Parler-TTS · [huggingface.co/ai4bharat](#)
- S16 Sarvam-30B open weights, Apache 2.0 · [huggingface.co/sarvamai](#)
- S17 E2E Networks: L4 ₹49/h, L40S ₹102/h · [e2enetworks.com](#)
- S18 Sarvam API pricing; carrier rate cards (team research)



Policy & our work

- S19 DPDP Rules 2025 (14 Nov 2025) · meity.gov.in
- S20 TRAI 1600 series for govt-to-citizen calls · traai.gov.in
- S21 Our work: prototype code, MODELS.md, cost_model.py, measure.py · [GitHub](https://github.com)

SEE MORE *code, prototype, video and every calculation, one scan each*



GitHub repo

code, architecture,
docs/MODELS.md

Try the prototype

call number + PIN (add)



Demo video

a real call, start to finish (add link)



Calculations

scripts/cost_model.py: every
rate + source